(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

PRACTICAL NO.01	6 HOURS

Practice Lab Assignments:

1. To accept an object mass in kilograms and velocity in meters per second and display its momentum. Momentum is calculated as e=mc2 where m is the mass of the object and c is its velocity.

2. Write a Python program for following conditions.

- If n is single digit print square of it.
- If n is two digit print square root of it.
- If n is three digit print cube root of it.

3. Read the birth date and salary in rupees of employees. Perform data transformation for birthdate to age and also salary which is in rupees to salary in dollars using functions.

4. Print the reverse number of a given number,

Momentum Calculation

```
# Momentum = m * c^2

m = float(input("Enter mass (kg): "))
c = float(input("Enter velocity (m/s): "))

momentum = m * (c ** 2)

print("Momentum =", momentum)
```

Output

```
Enter mass (kg): 2
Enter velocity (m/s): 6
Momentum = 72.0
```

Number Condition Program

```python
import math

n = int(input("Enter a number: "))

if 0 <= n <= 9:
    print("Square =", n ** 2)

elif 10 <= n <= 99:
    print("Square root =", math.sqrt(n))

elif 100 <= n <= 999:
    print("Cube root =", n ** (1/3))

else:
    print("Number not in valid range")
```

Output

```
Enter a number: 4
Square = 16
```

Employee Data Transformation (Using Functions)

```python
def calculate_age(birth_year):
    current_year = 2026
    return current_year - birth_year

def rupees_to_dollars(rupees):
    return rupees / 83   # approx conversion

birth_year = int(input("Enter birth year: "))
salary_rupees = float(input("Enter salary in rupees: "))

age = calculate_age(birth_year)
salary_dollars = rupees_to_dollars(salary_rupees)

print("Age =", age)
print("Salary in Dollars =", salary_dollars)
```

Output

```
Enter birth year: 2006
Enter salary in rupees: 50000
Age = 20
Salary in Dollars = 602.4096385542168
```

Reverse a Number

```python
num = int(input("Enter a number: "))

reverse = 0
while num > 0:
    digit = num % 10
    reverse = reverse * 10 + digit
    num = num // 10

print("Reversed Number =", reverse)
```

Output

```
Enter a number: 5
Reversed Number = 5
```